



Exploring the World of Science

Division C Rules Manual

Division C (Gr. 9-12)

SCIENCE OLYMPIAD NIGERIA © 2025



WELCOME TO THE 2026 SCIENCE OLYMPIAD!

This Rules Manual will help you prepare to compete in the National Tournament. Each Science Olympiad event has a corresponding page on the Science Olympiad national website. All users of this manual are subject to the Terms of Use Agreement. To compete, users must first join the Science Olympiad program in their home state and become registered members of their school team.

See our website for info on participation, Policies and Terms of Use at www.sciolyng.org

Division C (SSS 1 - 3) Participation Rules

A team may have up to twelve (12) members. A maximum of Six (6) SSS 3 students is permitted on a division C team, while other team members can be a combination of a minimum of two (2) SSS 1 and four (4) SSS 2 students.

Division B (JSS 1 - 3) Participation Rules

A team may have up to twelve (12) members. A maximum of six (6) JSS 3 students is permitted on a Division B team, then a combination of up to two (2) of JSS 1 and four (4) JSS 2 students to be part of the team.

Students Below Grade Level Designations

Science Olympiad encourages students to participate in the Division that matches current Science Olympiad grade level designations. However, to support the inclusion of students who wish to participate in Science Olympiad, schools with grade levels lower than those stated in a Division are permitted to invite members below the grade level designations. Participation is limited to age-appropriate events (as determined by a coach, principal or tournament director) and prohibited where safety is a concern (such as the use of chemicals).

Science Olympiad Team Membership

Science Olympiad requires that all teams (up to 15 members) competing in any Science Olympiad Tournament (Invitational, Regional, State or National) must be a member of Science Olympiad by the ir school's registration. There is no exception to this requirement, regardless of what teams from the same school are called (Varsity, JV, Alternate Team, Extra Team, Team Two, Team B). No school, region or state Science Olympiad organization is allowed to alter or amend these national membership requirements. Please see the Science Olympiad Copyrights and Use Statement outlining use of Science Olympiad Rules and procedures at sanctioned tournaments.

Find more Science Olympiad team information under the Policies section of the national website: Code of Ethics & Rules, Copyrights and Use, Building Policy, Eye Protection.



A Science Olympiad national tournament (preliminary to the national) typically consists of 23 - 24 different events, and those 23 events can be classified into one of four event types. This information is being provided so that Science Olympiad participants can more easily identify events that they may enjoy competing in, regardless of the event content. Coaches can approach coaching from the perspective of event type as opposed to event content, and teams can be aware of how the format of the tournament might affect available events. The symbol to the left of each description has been added to the upper right-hand corner of each Event Rule to identify the event type.



Core Knowledge Event: An event where participants are given a set of topics that they are expected to research and master the factual content. Mastery is demonstrated at a tournament by taking a paper-pen, station, and/or computer test.

Core Knowledge Events can be run regardless of the tournament format that has been chosen by the State Chapter and the Tournament Director.



Build Event: An event where participants are given some specifications about a device or object they are expected to design, create, and test in advance of the tournament. The devices or objects are often modified on site to account for an unknown parameter prior to testing or evaluation.

In some cases, Build Events may or may not be run depending upon the format of the Science Olympiad tournament being conducted. The Tournament Director will make these decisions to ensure safety and fairness for all teams. If a Build Event is not to be run at a tournament, the Tournament Director will notify all teams in advance of the given tournament.



Laboratory/Hands-On Event: An event where participants are given a general topic in which they will be expected to deepen their content knowledge of the topic and associated research techniques prior to the tournament. At the tournament they will be assessed by the completion of a hands-on task, which may or may not require a written report, within a defined timeframe.

Depending upon the format of Science Olympiad Tournament being held, there may be some alterations to or cancellation of Lab Events. To the greatest extent possible, Tournament Directors will work to ensure Lab Events are conducted, though that may mean hands-on activities are omitted and participants will work with previously collected data. The Tournament Director will make these decisions to ensure safety and fairness for all teams. If a Lab Event is altered or not to be run at a tournament, the Tournament Director will notify all teams in advance of the given tournament.



Hybrid Event: An event which combines elements from two or more of the above event types. The most common combination mixes elements of a Core Knowledge Event with elements of a Build or Lab Event.

As with the previous events, Hybrid Events may be altered to fit the format of the Science Olympiad Tournament being held. This may mean that Lab or Build elements of the event are modified or not conducted. The Tournament Director will make these decisions to ensure safety and fairness for all teams. If a Hybrid Event is altered or not to be run at a tournament, the Tournament Director will notify all teams in advance of the given tournament.

GENERAL RULES, CODE OF ETHICS, AND SPIRIT OF THE PROBLEM

The goal of competition is to give one's best effort while displaying honesty, integrity, and good sportsmanship. Everyone is expected to display courtesy and respect - see Science Olympiad Pledges. Teams are expected to make an honest effort to follow the rules and the spirit of the problem (not interpret the rules so they have an unfair advantage). Failure by a participant, coach, or guest to abide by these codes, accepted safety procedures, or rules below, may result in an assessment of penalty points or, in rare cases, disqualification by the tournament director from the event, the tournament, or future tournaments.

1. Actions and items (e.g., tools, notes, resources, supplies, electronics, etc.) are permitted, unless they are explicitly excluded in the rules, are unsafe, or violate the spirit of the problem.
2. While competing in an event, participants may not leave without the event supervisor's approval and must not receive any external assistance. All electronic devices capable of external communication as well as calculator applications on multipurpose devices (e.g., laptop, phone, tablet) are not permitted unless expressly permitted in the event rule or by an event supervisor. Cell phones, are not permitted.
3. Participants, coaches and other adults are responsible for ensuring that any applicable school or Science Olympiad policy, law, or regulation is not broken.
4. All pre-built devices presented for judging must be constructed, impounded, and operated by one or more of the 15 current team members unless stated otherwise in the rules. If a device has been removed from the event area, appeals related to that device will not be considered.
5. Officials are encouraged to apply the least restrictive penalty for rules infractions - see examples in the Scoring Guidelines. Event supervisors must provide prompt notification of any penalty, disqualification or tier ranking.
6. State and regional tournament directors must notify teams of any site-dependent rule or other rule modification with as much notice as possible.
7. At the discretion of the Tournament Director, portions of Hybrid Events containing hands-on activities as well as Build and Lab Events may be dropped from the tournament or be conducted as trial events.
8. At the discretion of the Tournament Director and Event Supervisors, completion time may be used as a tiebreaker for Core Knowledge and other events where a written or online test is used.



1. **DESCRIPTION:** Participants will be assessed on their understanding of the anatomy and physiology for the human Integumentary, Skeletal, and Muscular systems.

A TEAM OF UP TO: 2

APPROXIMATE TIME: 50 minutes

2. **EVENT PARAMETERS:**

Each team may bring one plain 8.5" x 11" sheet of paper, which may be in a sheet protector sealed by tape or laminated, that must not contain information on both sides in any form and from any source without any annotations or labels affixed.

3. **THE COMPETITION:**

Participants will complete a written test limited to the following topics.

a. **INTEGUMENTARY SYSTEM:**

- i. Functions of the Integumentary System
- ii. Anatomy of the layers of the skin, the component parts of the skin and sensory receptors
- iii. Skin Color and Texture, Hair and Nails, Integumentary Glands and the effects of aging on the skin
- iv. The diseases on each level from the cell to the whole person as listed: allergies to allergens (i.e., poison ivy, metals), infections (i.e., boils, carbuncles, athlete's foot, impetigo, **acne, human papilloma virus (HPV), other types of dermatitis**) and skin cancer
- v. **Injuries to the skin: burns, bedsores, calluses, and scars**
- vi. **National Tournament Only:**
 - (1) Additional disorders: Psoriasis, scabies, **ringworm, herpes, yeast infection, eczema & albinism**
 - (2) Treatments and/or prevention for all conditions listed above (drugs, surgery, etc.)
 - (3) **Aspects of wound healing including, but not limited to: inflammation, necrosis, apoptosis, vasodilation, and clotting**

b. **SKELETAL SYSTEM:**

- i. Bones of the axial and appendicular skeleton; label the basic surface anatomy of a bone as shown on a diagram and/or normal X-ray, CT and MRI
- ii. Name, structure and function of joint types and muscle and ligament attachments that surround the joints and the ranges of motion allowed by each type (e.g., ball and socket)
- iii. Structures of bones in cross-section
- iv. Cellular composition, structure and function of bones, bone marrow and cartilage
- v. Development and maturation of bones at the cellular and gross anatomical levels
- vi. How to distinguish between types of vertebrae (e.g., cervical, thoracic and lumbar)
- vii. Characteristics and radiological features of bone diseases/disorders from the cell level to the whole person as listed: osteoarthritis, osteoporosis, fractures, disc herniation, scoliosis, anterior cruciate ligament tears, medial collateral ligament damage, spinal fractures
- viii. The effects of exercise and aging on the skeletal system and the diseases mentioned above
- ix. Growth plate injuries - how they occur and the long-term effects

c. **MUSCULAR SYSTEM:**

- i. The interaction of the skeletal and muscular systems to allow movement
- ii. Muscle fibers - the cellular and gross anatomy of skeletal muscle, cardiac muscle & smooth muscle
- iii. Physiology of the skeletal muscle contraction system and the neuromuscular junction
- iv. How the skeletal muscles move bone, maintain posture, and produce heat
- v. Skeletal muscle actions – origin, insertion, interactions of different muscles
- vi. Cardiac and smooth muscle roles in the body
- vii. Location and identification, including origin, insertion, and function, of the major skeletal muscles of the body.



- viii. **The diseases on each level from cellular to the whole person: Muscular Dystrophy, Cerebral palsy, Poliomyelitis, Myasthenia gravis, tetanus, myositis**
- ix. Exercise and aging effects on the cellular and gross anatomical structures of the muscular system
- x. Muscle and tendon injuries and their prevention (i.e., strains and sprains)
- xi. Role of the nervous system in muscle function

4. SCORING:

- a. High score wins.
- b. Selected questions will be used to break ties.

2025 Science Olympiad Major Skeletal Muscles

Head and Neck

Frontalis
 Orbicularis oris
 Orbicularis oculi
 Occipitofrontalis
 Zygomaticus major
 Masseter
 Sternocleidomastoid
 Trapezius
 Buccinator

Move the Upper Extremities

Pectoralis major
 Biceps brachii
 Triceps brachii
 Flexor carpi radialis
 Flexor digitorum superficialis
 Extensor carpi radialis
 Extensor digitorum
 Extensor digiti minimi
 Extensor carpi ulnaris

Muscles of the Trunk

External Intercostals
 Internal Intercostals
 Transverse abdominis
 Rectus abdominis
 Serratus anterior
 Diaphragm

Move the Lower Extremities

Gluteus maximus
 Gluteus medius
 Tensor fasciae latae
 Semimembranosus
 Semitendinosus
 Biceps femoris
 Rectus femoris
 Tibialis anterior
 Gastrocnemius



1. **DESCRIPTION:** Teams will complete one or more tasks and answer a series of questions involving the science processes of chemistry focused in the areas of Aqueous Solutions and Acids & Bases.

A TEAM OF UP TO: 2

EYE PROTECTION: C

APPROXIMATE TIME: 50 minutes

2. **EVENT PARAMETERS:**

- a. Each participant must bring safety equipment (e.g., goggles, lab coat, apron), a writing implement, and may bring a stand-alone calculator of any type.
- b. Each participant may bring one 8.5" x 11" sheet of paper, which may be in a sheet protector sealed by tape or laminated, with no information on both sides in any form and from any source.
- c. Teams should bring any or all of the items listed as Recommended Lab Equipment for Division C Chemistry Events. Teams not bringing these items will be at a disadvantage, as they are not provided.
- d. Participants must wear goggles, an apron or a lab coat and have skin covered from the neck down to the wrist and toes. Gloves are optional, but if the host requires a specific type they will notify teams. Shoulder length or longer hair must be tied back. Participants removing safety clothing/goggles or unsafely handling materials or equipment will be penalized or disqualified.
- e. Supervisors will provide any required reagents, additional glassware, and/or references that are needed for the tasks (e.g., Periodic Table, table of standard reduction potentials, any constants needed).

3. **THE COMPETITION:**

- a. The competition will consist of a series of tasks similar to those in first year high school subjects. These tasks could include hands-on activities, questions on listed topics, interpretation of data (e.g., graphs, diagrams, tables), or observation of an established and running experiment.
- b. Nomenclature, formula writing, & stoichiometry (mole conversions & percentage yield) are essential tools of chemistry & may be included in the event. Participants are expected to know the symbols & charges for: nitrate, carbonate, phosphate, acetate, sulfate, ammonium, bicarbonate, & hydroxide. Participants should know how to use the "ite" form of anion (one less oxygen than the "ate" form). With a periodic table, participants should be able to obtain charges for monatomic ions (e.g., Na^+ , S^{2-}).
- c. Participants should understand the following Acid-Base Chemistry concepts:
 - i. Properties & Uses of Common Acids and Bases
 - (1) Acids - (HCl , HNO_3 , H_2SO_4 , H_3PO_4 , H_2CO_3 , acetic, and ascorbic acid)
 - (2) Bases - (NaOH , KOH , $\text{Ca}(\text{OH})_2$, $\text{Mg}(\text{OH})_2$, and $\text{NH}_3(\text{aq})$)
 - (3) Acid/Base indicators and how they are used; pH ranges and color changes will be provided. Questions will not address theories of how indicators work.
 - ii. Acid & Base reactions will be limited to metals, carbonates, bicarbonates, sulfites, bisulfites, oxides, & neutralizations
 - iii. Titrations to determine percent composition, molarity, and/or molecular mass
 - iv. Additional calculations will be limited to K_a , K_b , pH, pOH, and dilution
 - v. State and Nationals Only: calculations or questions about buffers
- d. Participants should understand the following about Aqueous Solutions:
 - i. Principles, properties, terms, and definitions concerning aqueous solutions
 - ii. Calculate solution concentration given quantities of solute and solvent
 - iii. Calculate the amount of material needed to achieve specific concentration
 - iv. Different measurements of concentration (e.g., molarity, molality, mass percentage, and parts per million) and how to calculate each
 - v. State and Nationals Only: conversions between concentration units

4. **SAMPLE QUESTIONS/ACTIVITIES:**

- a. Titrations to determine percent composition, molarity, and/or molecular mass.
- b. Given a pH indicator and the results of a test determine the pH of a solution.
- c. Identify the pH indicator that should be used to monitor the pH change in a given experiment.
- d. Use freezing point depression to determine the molar mass of a solute.
- e. Identify and explain factors that affect solution formation.
- f. Determine whether a solution is saturated, unsaturated, or supersaturated.

5. **SCORING:**

- a. High score wins. Points will be divided evenly between Aqueous Solutions and Acids & Bases.
- b. Time may be limited at each task but will not be used as a tiebreaker or for scoring.
- c. Ties will be broken by pre-selected questions.
- d. A penalty of up to 10% may be given if the area is not cleaned up as instructed.
- e. A penalty of up to 10% may be given if a team brings prohibited lab equipment to the event.



1. **DESCRIPTION:** Participants must complete tasks and answer questions about electricity and magnetism.

A TEAM OF UP TO: 2

EYE PROTECTION: None

APPROXIMATE TIME: 50 minutes

2. **EVENT PARAMETERS:**

- a. Each team may not bring any sheet or note of any size containing information in any form and from any source.
- b. Each team may bring writing utensils and stand-alone calculators of any type for use during any part of the event.
- c. Event Supervisors must provide any material & measurement devices required for the hands-on tasks.
- d. Participants may bring their own basic multimeters for use in place of provided ones at the discretion of the Event Supervisor.

3. **THE COMPETITION**

Part I: Written Test

- a. The written test consisting of multiple choice, true-false, completion, or calculation questions/problems will assess the team's knowledge of electricity and magnetism.
- b. Unless otherwise requested, answers must be in metric units with appropriate significant figures.
- c. The test will consist of at least one question from each of the following areas:
 - i. Historical perspective of the electricity discoveries made by Ampere, Coulomb, Kirchhoff, Volta, Ohm, Tesla, & Faraday
 - ii. Properties of electric charges/fields, sources/hazards of static electricity, Coulomb's Law, capacitance
 - iii. Direct current (DC) characteristics, sources, uses, simple circuit diagrams, DC hazards
 - iv. Alternating current (AC) characteristics, sources, uses, AC hazards
 - v. Concepts and units of current, voltage, resistance, power, energy, and using Ohm's law
 - vi. Magnetic poles/fields, electromagnets, transformers, motors/generators, right-hand rule
 - vii. Electrical control devices including 3-way light switch circuits
 - viii. Simple calculations, constructions, and configurations of a circuit and individual components
 - ix. Fundamental characteristics and operation of a light emitting diode (LED)
 - x. Simple circuit analysis using Kirchhoff's Voltage & Current Laws
 - xi. Division C only - Basic digital logic and digital logic operations
 - xii. Division C only - Basic electrical characteristics of silicon PN junctions (e.g., Diodes, PNP, NPN)
 - xiii. Division C only - Basics and applications of Operational Amplifiers (OpAmps)
- d. Topics not included in the competition are: semiconductors (beyond those listed above), AC circuit theory, inductance, calculations involving direct use of calculus and/or differential equations, non-linear devices, three-state logic gates, sequential logic, 3 Phase Power, and oscilloscopes.

Part II: Hands-On Tasks

- a. The hands-on portion will consist of at least one task at a station(s) for the teams to complete.
- b. Participants must be familiar with the operation of breadboards and how to use them. **Participants may ask event supervisors for details of the internal wiring of any breadboards used for the tasks.**
- c. The hands-on tasks, or stations, may include but are not limited to:
 - i. Determine the value of a mystery resistor in a circuit using only voltage measurements.
 - ii. Calculate the power supplied to a circuit.
 - iii. Given some wires, batteries, resistors, and 2 LEDs, hook them up so the LEDs are equally bright.
 - iv. Construct an electromagnet using some wire, a bolt and battery.

4. **SCORING:**

- a. High score wins.
- b. Points will be awarded for correct answers, measurements, calculations, and data analysis. Supervisors are encouraged to provide a standard form for competitors to show measurements/calculations.
- c. The written portion of the competition will account for 50-75% of each team's score. No single question will count for more than 10% of the total points possible on the written test.
- d. The hands-on portion of the competition will account for the remaining 25-50% of each team's score.
- e. Ties will be broken using pre-selected task(s)/question(s) that will be noted on the written test.



1. **DESCRIPTION:** Teams will complete tasks related to physical and geological oceanography.

A TEAM OF UP TO: 2

APPROXIMATE TIME: 50 minutes

2. **EVENT PARAMETERS:**

Each team may bring plain sheets of paper, sheet protector, containing no information in any form and from any source. If the event features a rotation through a series of laboratory stations where the participants interact with samples, specimens, or displays; no material may be removed from the binder throughout the event. All necessary writing utensils and implements may be brought as the Event Supervisor will not be providing them.

3. **THE COMPETITION:**

- a. Participants will be presented with questions which may include one or more tasks at a workstation or a timed station-to-station format.
- b. The participants will be expected to use process skills (e.g., communicating, classifying, inferring, measuring, observing, predicting, and using number relationships) to answer questions on the following topics:
 - i. Seawater: composition, density, variations in salinity, and sources of salts
 - ii. Shortwave and longwave radiation, sensible and latent heat fluxes, geothermal heat, and heat transport
 - iii. Water temperature, pressure, and the three-layer structure of ocean water
 - iv. Topographic features found at continental margins, estuaries, ocean basins, and mid-ocean ridges
 - v. Processes and features of tectonic plate motion in ocean basins and patterns of age of the ocean floor
 - vi. Distribution of chemicals (e.g., nutrients, oxygen, metals) in the ocean, as well as vertical and horizontal structure
 - vii. Formation of fringing reefs, barrier reefs, and atolls
 - viii. Waves: Motion, height, wavelength, period, fetch, swell, surf, and tsunamis
 - ix. Surface currents: Warm and cold currents, Coriolis effect, and gyres
 - x. **Division C Only:** Geostrophic balances
 - xi. Coastal currents: longshore currents, rip currents, and upwelling
 - xii. Deep ocean circulation, ocean overturning, and water masses
 - xiii. High and low tides, spring and neap tides, tidal currents, and tidal resonance
 - xiv. Coastal features and processes, uplift and subsidence, and influence on sea level rise
 - xv. Oceanic tools used for research (e.g., collection of water samples, sediments, cores, and tracking water movement)
 - xvi. Relationships between fisheries and ocean circulation (e.g., upwelling, El Niño, Pacific Decadal Oscillation)

4. **REPRESENTATIVE ACTIVITIES:**

- a. Given the water temperatures at various depths in a column of seawater, teams will construct graphs to identify and label the thermocline.
- b. Use a downloaded dataset of oxygen to identify water masses and pathways of circulation.
- c. Identify topographic features of ocean regions using seafloor maps.
- d. Write a hypothesis to explain changes in water salinity in high latitude ocean regions.
- e. Analyze and interpret water pH data from selected regions where barrier reef formation is changing.
- f. Given a set of vertical profiles of salinity in an estuary, identify the type of estuary (e.g., fjord, salt wedge, well-mixed, partially-mixed) and discuss implications for bottom water hypoxia.
- g. Relate trends in coastal flooding at specific locations to global sea level rise and local subsidence.

5. **SCORING:** Points will be awarded for the quality and accuracy of responses. High score wins. Ties will be broken by the accuracy and/or quality of answers to selected questions.



1. **DESCRIPTION:** Teams must design, build, and test one vehicle that uses electrical energy as its sole means of propulsion to travel as quickly as possible and stop close to a Target Point.

A TEAM OF UP TO:2

EYE PROTECTION: Yes

IMPOUND: Yes

TIME: 8 minutes

2. **CONSTRUCTION PARAMETERS:**

- a. The vehicle must be designed to travel between 9.00 and 12.00 meters and come to a complete stop. The exact distance in these intervals (0.10 m - states) must be chosen by the Event Supervisor (ES) and will not be announced until the impound period is over.
 - b. Electrical energy used within the vehicle for any purpose must be stored in common, commercially available batteries labeled with their voltage by the manufacturer. Lead acid batteries are not allowed.
 - c. The vehicle may use no more than 6 individual cells (labeled 1.5 volts or less each), a single 9-volt battery, or single battery pack (labeled 7.2 volts or less) at any given time during the 8 minutes. Batteries need not be installed until immediately prior to the run. Only electrical energy stored in these batteries may be used to propel the vehicle.
 - d. Additional energy storage devices may be used to operate other functions (e.g., braking system) provided these functions do NOT help propel the vehicle in any way. All sources of energy must be in an easily accessible locations for inspection by the ES.
 - e. Components (e.g., motors, gearboxes, motor controls, bodies, and chassis) may be purchased or made by the team members. Electronic components (i.e., solid-state devices such as transistors, integrated circuits, diodes, and microprocessors) are allowed.
 - f. Only the wheels or treads in the ready-to-run configuration must fit in a 35.0 cm x 60.0 cm space. There is no height restriction.
 - g. A single 14-inch wooden round dowel must be attached to the front end of the vehicle. The dowel must be approximately perpendicular to the floor and must be the leading part of the vehicle at all times.
 - i. The dowel must extend to at least 20.0 cm from the floor.
 - ii. The dowel must also extend to within 1.0 cm of the track's surface so that its front bottom edge will be the vehicle's Measurement Point for distance measurements.
 - iii. No part of the vehicle, except for a dowel attachment device of no more than 2.0 cm, can extend beyond the front of the 14-inch dowel during its entire run.
 - h. Competitors must start the vehicle by using any part of an unsharpened pencil with an unused eraser (provided by the ES) to actuate a button or switch by pushing it perpendicular to the floor (vertical). A nonvertical actuation is a construction violation.
 - i. Sighting, aligning, and guiding devices are allowed, including those that use electricity.
 - j. The only parts of the vehicle that are allowed to contact the floor during the run are the wheels/treads and those that are already in contact with the floor in the ready-to-run position. Piece(s) falling from the vehicle result in a construction violation.
 - k. The vehicle must not be remotely controlled or tethered. The stopping mechanism must work automatically.
1. Competitors must be able to answer questions regarding the design, construction, and operation of the device per the Building Policy.

3. **THE TRACK:** At the Event Supervisor's discretion, more than one track may be used. Teams must be given the option to choose which track they will use. Both runs by a team must be made on the same track.
 - a. The track must be on a smooth, level, hard surface.
 - b. The ES must use tape approx. 2.5 cm wide and 1.00 m long to mark the track's Starting Line, Target Line, and the lines for timing (0.5 m and 8.5 m from the Starting Line). The Starting Point and Target Point, located at the center of the Starting Line and Target Line will be clearly marked.



4. **THE COMPETITION:** Teams filing an appeal must leave all impounded material with the Supervisor.
 - a. The vehicle, batteries, and spare parts must be impounded before the start of the competition. Tools, data and calculating devices need not be impounded
 - b. Only competitors and supervisors must be allowed in the impound and track areas during the event. Once competitors enter the event area, they must not leave the area or receive outside assistance, materials, or communication.
 - c. Teams must be given 8 minutes to set up their vehicle and complete up to 2 runs. Vehicles in the ready-to run configuration that starts before the end of the 8-minute time period will be allowed to complete a run.
 - d. In the ready-to-run configuration, teams must place the vehicle anywhere behind the Starting Line. No part of the vehicle can be further than 1.00 m behind the Starting Line. The vehicle must remain behind the Starting Line without being touched.
 - e. Teams may adjust their vehicle before each run (e.g., change its speed, distance, directional control, change batteries) within their 8 minutes providing the vehicle continues to meet specifications.
 - f. Teams may use their own measuring devices to verify the track dimensions during their 8 minutes. They may not roll the vehicle on or adjacent to the track surface between the Starting and Target lines at any time prior to or during the competition.
 - g. Substances that may damage the floor or interfere with subsequent runs must not be applied to the wheels, treads, or floor. During their time, competitors may clean the track, but it must remain dry at all times.
 - h. Sighting, aiming and guiding devices placed on the track are permitted but must be removed before the vehicle begins its run. Aligning and sighting devices mounted on the vehicle may be removed at the team's discretion prior to each run.
 - i. If the vehicle does not move upon actuation, it does not count as a run, and the team may request to set up for another run but must not be given additional time.
 - j. Run time starts when the dowel of the vehicle reaches the 0.50 m Line and ends when it passes the 8.50 m Line. The Run Time must be recorded in seconds to the precision of the timing device used.
 - k. Once the vehicle starts a run, the competitors must not follow it but must wait until called by the ES.
 - l. The team's 8-minute time is paused when the vehicle stops to allow for the supervisor's measurements. Timing resumes once the competitors pick up their vehicle or begin making their own measurements.
 - m. If the dowel of the vehicle passes the 0.50 m Line but stops before the 8.50 m Line, the timers record the Run Time, which stops when the vehicle comes to a complete stop, and the run is scored with a Competition Violation
 - n. A Failed Run occurs if a second run does not occur in the 8 minutes or if the time or distance cannot be measured for a vehicle (e.g., it starts before the ES is ready, if it moves but does not go at least 0.50 m, or the competitors pick it up before it is measured).
5. **SCORING:** Low Final Score wins. The Final Score must be the lower of the 2 Run Scores. The Final Score of a vehicle with one Failed Run must be the other Run Score.
 - a. Run Score for each run = Time Score + Distance Score + Penalties
 - b. Time Score = Run Time x 10
 - c. Distance Score = the distance from the Measurement Point to the Target Point in centimeters measured to the nearest 0.1 cm, a point-to-point measurement. The vehicle may stop on either side of the Target Line.
 - d. Each Competition violation must incur 1000 pts penalty per occurrence for that run (max. of 4000 pts).
 - e. Each Construction Parameters violation must incur 5000 pts penalty for that run (max. of 15000 pts).
 - f. A vehicle not impounded during the impound period must incur 10000 pts penalty for each run.
 - g. Ties must be broken by this sequence:
 1. Lower non-used Run Score
 2. Lower Time Score of the betterer-run
 3. Lower Distance Score of the better.



ELECTRIC VEHICLE (CONT)



- h. Teams whose vehicle cannot complete any run within the 8 minutes or those who have two Failed Runs must be given participation points only.

Scoring Example: On its first run, a vehicle stopped 135.8 cm from the Target Point with a Run Time of 6.67 s. On its second run, the vehicle stopped 21.3 cm from the Target Point with a Run Time of 5.85 s and incurred a Competition Penalty of 1000 points. Run #1 Score: $(6.67 \times 10) + 135.8 + 0 = 202.5$ and Run #2 Score: $(5.85 \times 10) + 21.3 + 1000 = 1079.8$ Final Score: 202.5



1. **DESCRIPTION:** Teams must construct a collecting device prior to the tournament that is designed to collect heat and complete a written test on alternative energy concepts.

A TEAM OF UP TO: 2

IMPOUND: No

APPROX. TIME: 50 minutes

2. **EVENT PARAMETERS:**

- a. ~~Each team may bring their~~ heat collection device, an unaltered, glass or plastic, standard (height ~1.4 times the diameter) 250 mL beaker, copies of graphs and/or tables for scoring, tools, supplies, writing utensils, and a stand-alone calculator of any type for use during any part of the event.
- b. Event supervisors will supply water and thermometers. Non-contact thermometers are allowed.
- c. Participants must be able to answer questions regarding the design, construction, and operation of the device per the Building Policy.

3. **CONSTRUCTION PARAMETERS:**

- a. Devices may be constructed of and contain any materials (e.g., cardboard, aluminum foil, reflective fabric or material, glue, tape, mirrors, tiles and lenses).
- b. The device, including beaker, must fit within a 35.0 cm x 35.0 cm x 35.0 cm cube when set up for testing.
- c. Within the device, participants must be able to insert and remove a beaker that they supply (see 2.b).
- d. The device must also easily accommodate the insertion and removal of a thermometer into the beaker. Parts of the device may be inside the beaker, but the device must not contact the water.
- e. Devices will be inspected to ensure that there are no energy sources (e.g., no electrical components, small battery powered heaters, chemical reactions, etc.) to help warm the water. At the event supervisor's discretion, teams must disassemble their devices at the end of the testing period in order to verify the materials used in construction.
- f. All parts of the device must not be significantly different from room temperature at the start of the event.
- g. Prior to competition, teams must calibrate devices by preparing graphs/tables showing the relationship between elapsed time and water temperature. A labeled device diagram should be included.
 - i. Any number of graphs and/or data tables may be submitted but the team must indicate up to four to be used for the Chart Score, otherwise the first four provided are scored.
 - ii. Graphs and/or tables may be computer generated or drawn by hand on graph paper. Each data series counts as a separate graph.
 - iii. Teams are encouraged to have a duplicate set to use, as those submitted may not be returned.

4. **THE COMPETITION:**

Part I: Written Test

- a. Teams will be given a minimum of 20 minutes to complete a written test consisting of multiple choice, true-false, completion, or calculation questions/problems.
- b. Unless otherwise requested, answers must be in metric units with appropriate significant figures.
- c. The competition must consist of at least five questions from each of the following areas:
 - i. Basic information and definitions about energy, work, heat and heat transfer, temperature, temperature scales, thermal energy and insulation.
 - ii. General information about renewable energy including but not limited to solar, wind, hydroelectric, tidal, ocean thermal energy conversion (OTEC), and geothermal.
 - iii. General information about energy conservation practices including but not limited to recycling, reusing, and using materials with greater efficiency.
 - iv. Mathematical relationships and equations used in determining heat loss and gain, specific heat, and heat transfer.

Part II: Device Testing

- a. At the start of the competition block, teams will be given 5 minutes to set up or modify their devices and use their graphs and/or tables to calibrate them. Devices that do not meet the construction specs will not be allowed to be tested until brought into specification.



- b. At each station, the event supervisor will provide an incandescent lamp with a bell-shaped reflector. The lamp will be mounted, facing down, above the testing surface (on which teams will set up their device) such that the bottom of the bulb is at least 40.0 cm from the testing surface. Multiple identical stations may be used.
 - c. At the start of a team's device testing period the supervisor, using their own measuring device, will dispense 100 mL of water into the team's beaker. A team may elect to install the beaker in a device prior to this, but must leave sufficient access to the beaker. Otherwise the team may then place the beaker into their device.
 - d. Teams will use their graphs and/or tables to predict the temperature of the water in their beaker at the end of the 10-minute heating time. After receiving water, teams will be given at least 3, but no more than 5 minutes to make their final predictions. During this time, teams may use their own thermometers to measure the starting water temperature in their beaker, but after this time must remove them.
 - e. The supervisor will insert a probe/digital thermometer into the water to measure and record the initial temperature to the nearest tenth of a degree. Supervisors may leave thermometers/probes in the devices for the entire heating period, but will announce if they will do so before impound. Otherwise they will insert a thermometer/probe into the beaker in the device, wait at least 20 seconds, and record the resulting temperature. Multiple thermometers/probes may be used at the supervisor's discretion.
 - f. The light source must be turned on and a stopwatch started. At the end of 10 minutes the light will be turned off and the thermometer/probe will be read and recorded to the nearest tenth of a degree to determine the gain in temperature.
 - g. The supervisor will review with the team the Part II data recorded on their scoresheet.
 - h. Teams filing an appeal regarding Part II must leave their device in the competition area.
5. **SCORING:**
- a. High score wins.
 - b. All scoring calculations are to be done in degrees Celsius.
 - c. Final Score (FS) = TS + CS + HS + PS; The maximum possible FS is 100 points. A scoring spreadsheet is available at www.soinc.org.
 - d. Test Score (TS) = (Part I score / Highest Part I score for all teams) x 50 points
 - e. Chart Score (CS): One of the submitted graphs/tables, selected by the Event Supervisor, is scored using i., ii., and iii., described below for a maximum of 6 points. Four (4) additional CS points are available via items iv. and v. Partial credit may be given. A device must be present to receive a CS.
 - i. 2 points for including data spanning at least one variable range
 - ii. 2 points for including at least 10 data points
 - iii. 2 points for proper labeling (e.g., title, team name, units)
 - iv. 0.5 points for each distinct graph or table turned in (up to 2 points total)
 - v. 2 points for including a labeled device diagram
 - f. Heat Score (HS) = (HRF / Highest HRF of all teams) x 15 points; HRF (Heat Retention Factor) = (final beaker water temp / starting beaker water temp)
 - g. Prediction Score (PS) = (PE / Highest PE of all teams) x 25 points; PE (Prediction Estimate) = (1-(abs (final beaker water temp - predicted final beaker water temp) / final beaker water temp)). The minimum PS possible is 0 points.
 - h. If a team violates any COMPETITION rules, their HRF and PE values will be multiplied by 0.9 when calculating the scores.
 - i. If any CONSTRUCTION violation(s) are corrected during the Part II testing period the HRF and PE values will be multiplied by 0.7 when calculating the scores.
 - j. Teams that are disqualified for unsafe operation or do not bring a collecting device receive zero points for their HRF and PE scores. Teams will be allowed to compete in Part I.
 - k. Tie Breakers: 1st — Best TS; 2nd — Best HS; 3rd — Best PS

CHEMISTRY RECOMMENDED LAB EQUIPMENT

Each team may bring any or all of the items listed below for use in Division C Chemistry Events requiring laboratory equipment. Teams not bringing these items may be at a disadvantage as Event Supervisors will not provide Recommended Lab Equipment. A penalty of up to 10% may be given if a team brings prohibited lab equipment to the event.

Item & Expected Use	Likely to be used in:			
	Chemistry Lab	Forensics	Environmental Chemistry	Materials Science
Box - Containing all of the kit materials	X	X	X	X
10 ml Graduated Cylinder - Measuring volumes	X		X	
25 ml Graduated Cylinder - Measuring volumes	X		X	
100 ml Graduated Cylinder - Measuring volumes	X		X	
50 ml Beakers - Doing reactions, developing chromatograms	X	X	X	X
100 ml Beakers - Doing reactions, developing chromatograms	X	X	X	X
250 ml Beakers - Doing reactions, developing chromatograms	X	X	X	X
400 ml Beakers - Doing reactions, developing chromatograms	X	X	X	X
50 ml Erlenmeyer Flasks - Doing reactions	X		X	
125 ml Erlenmeyer Flasks - Doing reactions	X		X	
250 ml Erlenmeyer Flasks - Doing reactions	X		X	
Test Tubes - Mix Chemicals, heat chemicals	X	X	X	X
Test Tube Brush - Clean Test Tubes	X	X	X	X
Test Tube Holder - Holds test tubes for heating	X	X	X	
Test Tube Rack - Hold Test Tubes	X	X	X	X
Spot Plates - For semi-micro scale reactions, testing solubility, pH	X	X	X	
Petri Dishes - Doing reactions, developing chromatograms	X	X	X	X
Slides - To put hairs, crystals, or fibers on for use with a microscope		X		
Cover Slips - To cover & prevent items from coming off slides		X		
Droppers - Add small amounts of liquids to reactions	X	X	X	X
Spatulas or spoons - Getting small amounts of solids out of containers	X	X	X	X
Metal Tongs, Forceps, or Tweezers - Holding & retrieving objects	X	X	X	X
Stirring Rods - Stirring mixtures	X	X	X	X
Thermometer - Determining the temperature of a solution	X	X	X	
pH or Litmus paper - Test acidity or alkalinity of solution	X	X	X	
Hand Lens - Magnification of small items for identification		X		
Flame Loop - For identification of ions in a compound		X		
Cobalt Blue Glass - To filter out any sodium that might contaminate flame test from hands		X		
Filter Paper - Filter solids from liquids	X		X	
Funnel - Hold Filter Paper	X		X	
9V battery - Electrolysis	X		X	X
Alligator Clip Wires - Connecting meters to metals	X		X	X
Nail - Electrolysis	X		X	X
Piece of Cu metal - Electrolysis	X		X	X
Piece of Zn metal - Electrolysis	X		X	X
Multimeter - Measuring current, voltage, and resistivity	X		X	X
9V or less Battery Conductivity Tester - Determining ionic strength of solution	X	X	X	X
Calipers-mechanical, not digital - Measuring lengths very precisely	X			X
Paper Towels - Cleaning	X	X	X	X
Pencil - Writing, Marking Chromatogram		X		
Ruler - Measuring lengths		X		
Magnets - For extraction and identification of iron filings	X	X	X	X

The following document was prepared to offer some guidance to teams as they select calculators for use in different Science Olympiad events. By no means are the calculators listed here inclusive of all possible calculators; instead they are offered as common examples. The decisions of the event supervisors will be final.

Class I - Stand-alone non-graphing, non-programmable, non-scientific 4-function or 5-function calculators

are the most basic type of calculators and often look like the one shown to the right. These calculators are limited to the four basic mathematics functions and sometimes square roots. These calculators can often be found at dollar stores.

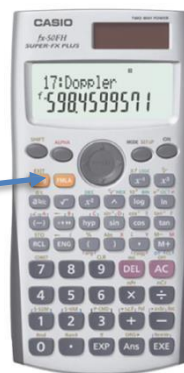


Class II - Stand-alone non-programmable, non-graphing calculators look like the calculator to the right or simpler. There are hundreds of calculators in this category but some common examples include: CASIO FX-260, Sharp EL-501, and TI-30X.



Class III- Stand-alone, programmable, graphing calculators and **stand-alone non-graphing, programmable calculators**, often look like the calculator shown on the right. Some examples are: Casio 975 0/9850/9860, HP 40/50/PRIME, and TI 83/84/89/NSPIRE/VOYAGE.

To identify a stand-alone non-graphing, programmable calculators are look for the presence of the 'EXE' button, the 'Prog' button, or a 'file' button. Examples include but are not limited to: Casio Super FXs, numerous older Casio models, and HP 35S. A calculator of this type with the buttons labeled is shown to the right.



PROG Button

EXE Button



Class IV - Calculator applications on multipurpose devices (e.g., laptop, phone, tablet, watch) are not allowed unless expressly permitted in the event rule.



Participant/Coach Responsibilities: Participants are responsible for providing their own protective eyewear. Science Olympiad is unable to determine the degree of hazard presented by equipment, materials and devices brought by the teams. Coaches must ensure the eye protection participants bring is adequate for the hazard. All protective eyewear must bear the manufacturer's mark Z87. At a tournament, teams without adequate eye protection will be given a chance to obtain eye protection if their assigned time permits. If required by the event, participants will not be allowed to compete without adequate eye protection. This is **non-negotiable**.

Corresponding Standards: Protective eyewear used in Science Olympiad must be manufactured to meet the American National Standards Institute (ANSI) standard applicable at its time of manufacture. The current standard is ANSI/ISEA Z87.1-2015. Competitors, coaches and event supervisors are not required to acquire a copy of the standard. The information in this document is sufficient to comply with current standards. Water is not a hazardous liquid and its use does not require protective eyewear unless it is under pressure or substances that create a hazard are added.

Compliant Eyewear Categories: If an event requires eye protection, the rules will identify one of these three categories. Compliance is simple as ABC:

CATEGORY A

- Description: Non-impact protection. They provide basic particle protection only
- Corresponding ANSI designation/required marking: Z87
- Examples: Safety glasses; Safety spectacles with side shields; and Particle protection goggles (these seal tightly to the face completely around the eyes and have direct vents around the sides, consisting of several small holes or a screen that can be seen through in a straight line)

CATEGORY B

- Description: Impact protection. They provide protection from a high inertia particle hazard (high mass or velocity)
- Corresponding ANSI designation/required marking: Z87+
- Example: High impact safety goggles

CATEGORY C

- Description: Indirect vent chemical/splash protection goggles. These seal tightly to the face completely around the eyes and have indirect vents constructed so that liquids do not have a direct path into the eye (or no vents at all). If you are able to see through the vent holes from one side to the other, they are NOT indirect vents
- Corresponding ANSI designation/required marking: Z87 (followed by D3 is the most modern designation but, it is not a requirement)
- Example: Indirect vent chemical/splash protection goggles

Examples of Non-Compliant Eyewear:

- Face shields/visors are secondary protective devices and are not approved in lieu of the primary eye protection devices below regardless of the type of vents they have.
- Prescription Glasses containing safety glass should not be confused with safety spectacles. "Safety glass" indicates the glass is made to minimize shattering when it breaks. Unless these glasses bear the Z87 mark they are not approved for use.

Notes:

1. A goggle that bears the Z87+ mark and is an indirect vent chemical/splash protection goggle will qualify for all three Categories A, B & C
2. VisorGogs do not seal completely to the face, but are acceptable as indirect vent chemical/splash protection goggles



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